

CHRISTCHURCH, New Zealand

WMO: 937810

Lat: 43.4897S

Lon: 172.5269E

Elev: 36

StdP: 100.90

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.8	-1.8	-4.2	2.7	1.5	-3.2	2.9	2.1	11.4	8.9	10.1	9.1	1.1	290	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	28.2	16.9	26.1	16.1	24.1	15.5	18.5	24.6	17.7	23.1	16.9	21.7	6.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.0	20.0	15.9	11.4	18.9	15.1	10.8	18.2	52.6	24.6	50.0	23.3	47.6	21.9	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	10.0	8.9			-4.8	32.5	0.8	1.5	-5.4	33.5	-5.9	34.4	-6.3	35.2	-6.9	36.3
			DB		-5.1	20.5	0.8	0.9	-5.6	21.2	-6.1	21.7	-6.5	22.2	-7.1	22.9
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	16.6	16.5	14.8	11.8	9.5	6.6	6.1	7.3	9.3	11.0	13.0	15.5
	DBStd	4.71	3.37	3.30	3.10	2.91	2.93	2.72	2.56	2.57	2.81	2.87	3.30	3.18
	HDD10.0	461	1	2	2	12	45	109	124	91	46	21	8	1
	HDD18.3	2566	76	69	117	196	274	352	378	342	272	227	163	99
	CDD10.0	997	205	182	151	67	29	6	4	7	24	53	97	172
	CDD18.3	61	21	17	7	1	0	0	0	0	0	0	3	12
	CDH23.3	604	191	152	78	10	1	0	0	0	0	6	40	126
	CDH26.7	140	52	43	14	0	0	0	0	0	0	0	6	25
Wind	WSAvg	4.0	4.8	4.6	4.2	3.5	3.3	3.0	3.1	3.5	3.9	4.4	4.6	4.8
Precipitation	PrecAvg	613	44	38	51	51	61	54	67	64	42	45	48	46
	PrecMax	913	139	115	173	199	164	168	181	171	111	138	117	154
	PrecMin	308	8	5	3	7	13	5	1	12	2	3	10	1
	PrecStd	133	31	23	40	43	33	35	47	41	30	28	26	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.9	31.0	28.8	24.9	22.0	18.1	17.8	18.9	21.9	24.1	27.1	29.2
		MCWB	17.6	18.0	17.0	14.6	13.9	11.4	11.2	11.4	12.9	14.7	16.1	16.9
	2%	DB	28.0	27.8	25.8	21.9	19.2	15.8	15.0	16.8	19.2	21.8	24.1	26.8
		MCWB	16.8	17.3	16.2	14.0	12.5	10.4	9.6	10.4	11.8	13.5	14.7	16.2
	5%	DB	25.2	24.9	22.9	19.2	17.1	13.8	13.0	14.8	17.2	19.2	22.0	24.2
		MCWB	16.0	16.5	15.2	13.1	11.7	9.6	8.6	9.4	11.0	12.5	14.1	15.4
	10%	DB	22.9	22.1	20.5	17.2	15.0	11.9	11.1	12.7	15.2	17.2	19.8	22.0
		MCWB	15.8	16.0	14.8	12.7	10.9	8.9	8.1	8.7	10.0	11.6	13.2	14.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	19.6	18.7	16.0	14.8	12.3	11.5	12.0	14.0	15.7	17.2	19.1
		MCDB	26.0	26.1	24.9	20.9	20.0	16.1	16.2	17.2	19.8	21.8	23.8	26.0
	2%	WB	18.5	18.4	17.4	14.9	13.5	11.0	10.2	10.7	12.6	14.1	15.9	17.6
		MCDB	24.2	24.4	22.6	19.3	17.3	14.2	13.0	14.9	17.9	19.9	22.0	23.6
	5%	WB	17.5	17.5	16.3	14.1	12.5	10.1	9.4	10.0	11.5	13.0	14.8	16.5
		MCDB	22.6	22.5	20.4	17.9	15.8	12.4	11.8	13.2	16.1	18.1	20.3	21.8
	10%	WB	16.7	16.7	15.5	13.2	11.6	9.3	8.5	9.2	10.6	12.1	13.8	15.6
		MCDB	21.0	20.7	19.1	16.5	14.5	11.3	10.6	11.7	14.2	16.5	18.6	20.1
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.0	10.3	10.1	9.8	9.6	9.5	9.7	10.5	10.5	10.8	10.1
		MCDBR	15.0	14.8	14.8	14.1	13.3	12.7	12.7	13.8	14.3	14.6	14.7	14.6
		MCWBR	6.2	6.1	6.8	7.7	7.9	8.4	8.3	8.4	8.2	8.0	7.2	6.4
	5% WB	MCDBR	11.5	11.9	11.4	11.5	11.1	10.4	10.5	11.1	12.3	12.7	13.0	11.8
		MCWBR	5.4	5.5	5.9	6.8	7.1	7.2	7.3	7.3	7.6	7.5	6.9	5.8
Clear-Sky Solar Irradiance	taub	0.334	0.327	0.311	0.301	0.284	0.276	0.275	0.289	0.305	0.317	0.319	0.328	
	taud	2.527	2.566	2.602	2.611	2.629	2.611	2.626	2.604	2.529	2.520	2.538	2.553	
	Ebn at Noon	990	971	941	879	826	795	828	880	931	970	1000	1003	
	Edh at Noon	108	98	86	72	59	54	58	71	90	101	106	106	
All-Sky Solar Radiation	RadAvg	6.30	5.64	4.32	2.81	1.84	1.41	1.68	2.43	3.67	5.06	6.17	6.39	
	RadStd	0.54	0.53	0.19	0.28	0.10	0.07	0.12	0.15	0.22	0.35	0.41	0.49	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	+0.33	+0.24	+0.41	+0.74	N/A	+0.25	-40	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	+0.31	N/A	N/A	+0.30	-39	N/A	N/A	N/A	N/A

Nomenclature: See separate page